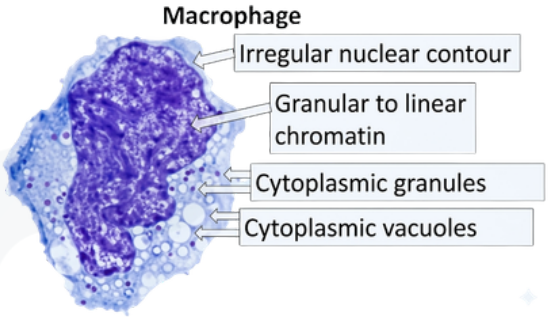
MACROPHAGES

Recall- Macrophages:

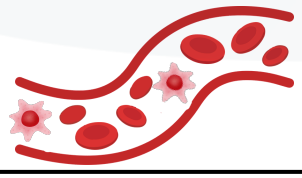
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Activation of Macrophages:

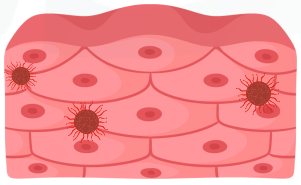
- The mononuclear phagocytic system consists of :

Monocytes circulating in the blood



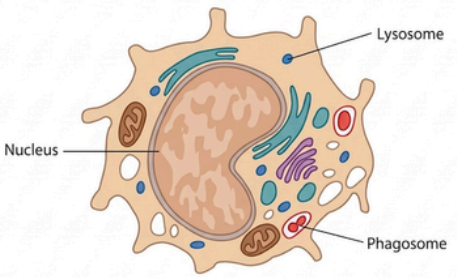
and

Macrophages in the tissues

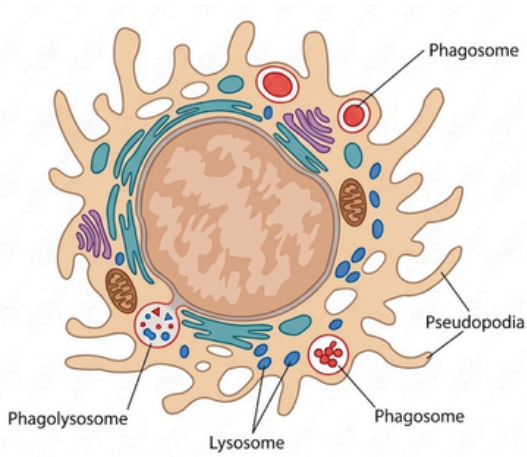


- Monocytes **circulate** in the bloodstream for about **8 h**, during which they **enlarge**; and then **migrate** into the tissues to **differentiate** into specific tissue macrophages
- Differentiation of a monocyte into a tissue macrophage allows it to:
  1. Become larger
  2. Increase its intracellular organelles in both number and complexity
  3. Acquire increased phagocytic ability producing higher levels of hydrolytic enzymes
  4. Ability to secrete variety of soluble factors (cytokines and chemokines)

Monocyte



Macrophage



NB

Macrophages have the dual capability of roaming freely in tissue or getting fixed in a particular tissue region

- Macrophages are activated by a variety of stimuli (such as bacteria, virus, parasites, excess immune complexes etc.)
- **Once activated,** macrophages exhibit **increased:**

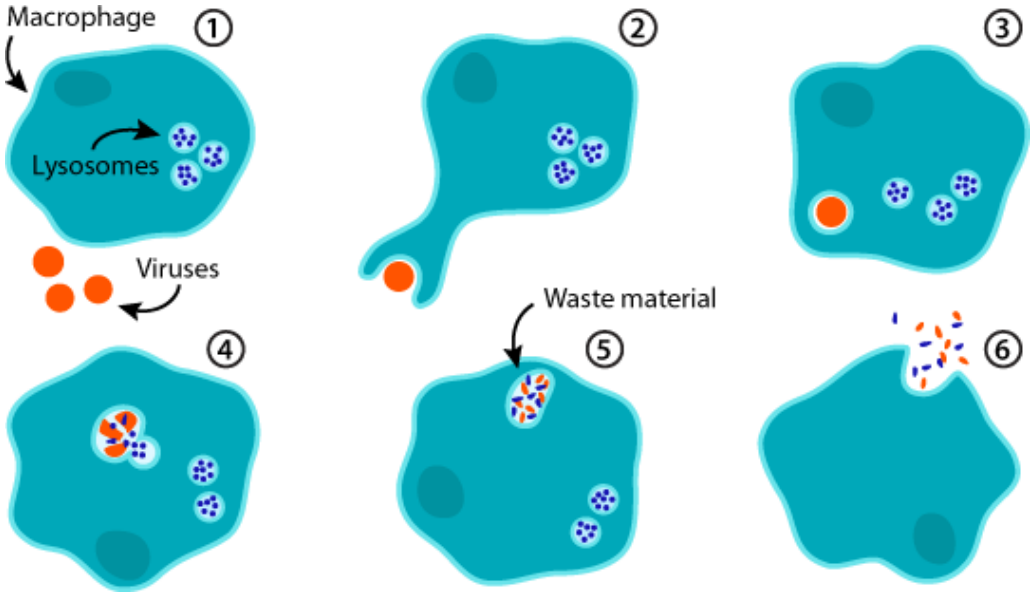
  1. phagocytic activity
  2. ability to kill ingested microbes
  3. secretion of inflammatory mediators, such as cytokines and chemokines — for communication and movement of immune cells
  4. ability secrete various **cytotoxic proteins** that help to eliminate a broad range of pathogens, including virus-infected cells, tumor cells, and intracellular bacteria

- **Main function** of macrophages is to perform **phagocytosis (engulf and kill)**

  Macrophages are capable of ingesting and digesting:

  1. Exogenous (foreign/external) antigens such as whole microorganisms and insoluble particles and;

  2. Endogenous (internal or host related) matter, such as injured or dead host cells, cellular debris, and activated clotting factors



IMPORTANT

Macrophages perform phagocytosis in the following manner:

Step 1: macrophages are attracted by the foreign agent/antigen such as bacteria and move toward it with the help of soluble substances such as chemokines and cytokines. This process is called **chemotaxis**

Step 2: macrophages **attach** to the antigen (infectious agent) through their cell membrane receptors

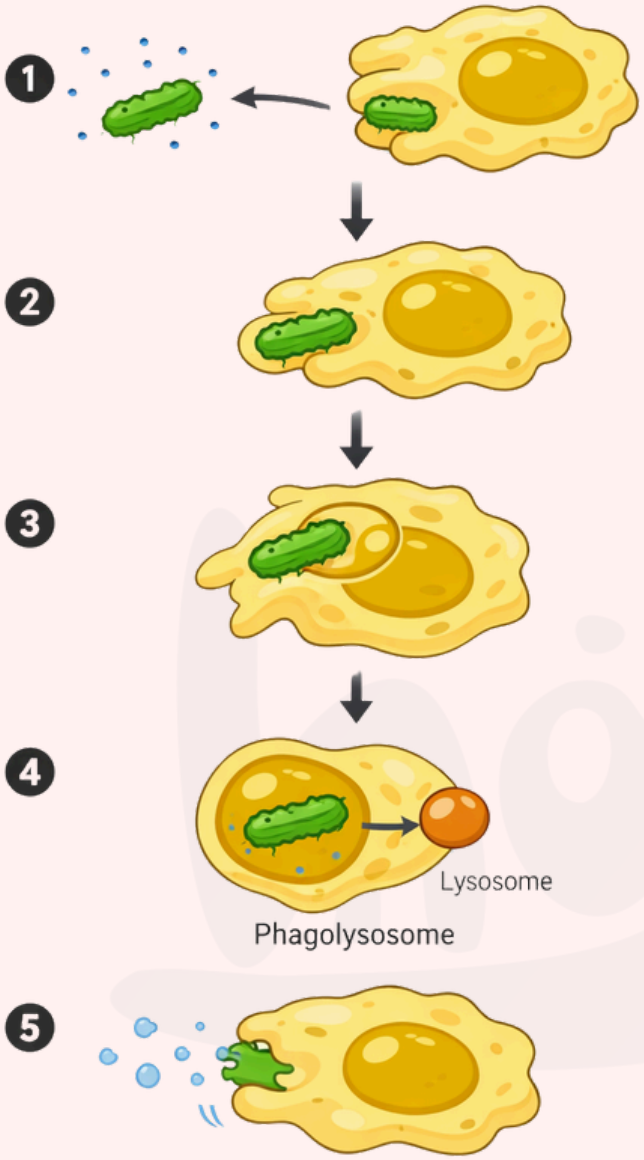
Step 3: Attachment induces membrane protrusions, called pseudopodia, to extend around the attached material
Fusion of the pseudopodia encloses the material within a membrane-bounded structure called a phagosome, which then enters the endocytic (internal killing) processing pathway

Step 4: In this pathway, a phagosome moves toward the cell interior, where it fuses with a lysosome to form a **phagolysosome**. The lysozyme and other hydrolytic enzymes digest the ingested material

Step 5: The digested contents of the phagolysosome are then eliminated in a process called **exocytosis**

NB

MACROPHAGE PHAGOCYTOSIS PROCESS



Chemotaxis

Attachment (Adherence)

Engulfment (Phagosome formation)

Fusion & Digestion (Phagolysosome formation)

Exocytosis

DESIGNS

Project Sample

Designed by **Dr. Abdurrahman Mahmoud Hafez**, Medical Content Designer.

Items Guide

Titles of the topic

Titles

- Definition
- LM
- EM
- Molecular structure
- Functions

Remember box

Remember

A point mentioned in other topic in the book and may cause some confusion

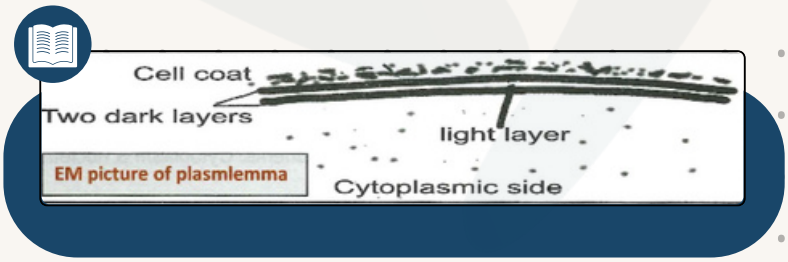
Mnemonics

Mnemonic

Important parts

IMPORTANT

Images from the handout



Additional images for Understanding



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Abdurrahman ® Hafez

Canva

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احترافي لكتابك

مع التخصص في
المحتوى الطبي



Dr: Abdurrahman M. **Hafez**

Faculty of Medicine, Kasr Al-Ainy - Cairo University

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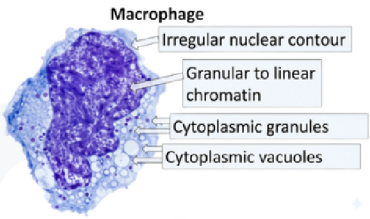
تصميم وتنسيق محتوى طبي



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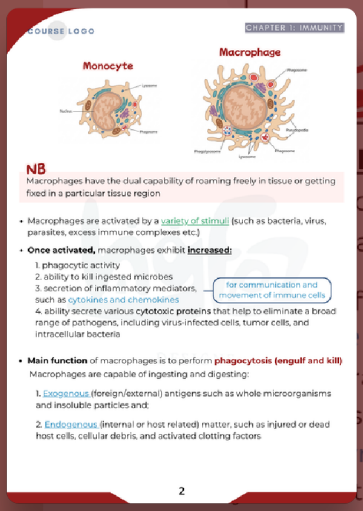
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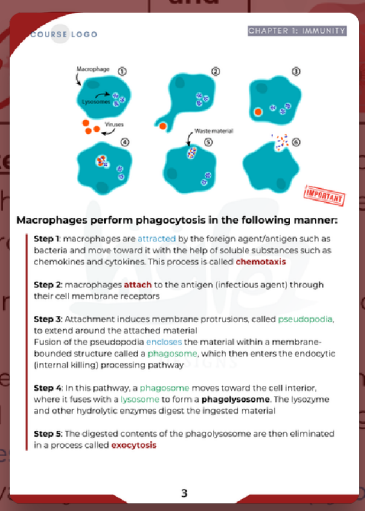


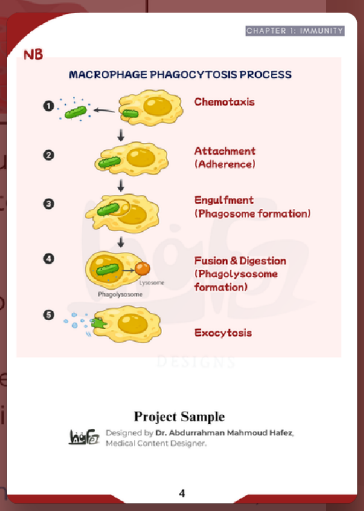
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1

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